

The issue was a one-time coupon, which because of the bug, could be used multiple times by a single customer. Due to this one major lapse the client lost millions of dollars in the space of a few hours. It escalated this issue to our company, and charged us thousands of dollars in fines. This is how lack of testing can result in huge losses.

Are mobile and Web testing the same?

The answer is certainly *no*. The most obvious difference between mobile app testing and Web app testing is that mobile apps are usually meant for a larger number of users, more diverse devices, and an increasingly broader range of communication than Web apps. The complexity level is more in mobile apps as compared to Web apps. In fact, the former are continually becoming increasingly complex, as they deliver more and more functions on mobile devices. A mobile application may run on several devices, including a smartphone, tablet, smartwatch, heart pacemaker, lock system, or fitness tracker.

The differences between mobile and Web apps

Devices and OSs: Desktop apps are usually targeted at specific platforms. Web based applications make it a bit more challenging because of the presence of browsers. Mobile applications, on the other hand, take the complexity of supported platforms to the next level because of the addition of devices. Mobile apps need to work on all types of devices (smartphones, tablets, and phablets) from many companies like Samsung, Sony, Nokia, HTC, Apple, etc – all fielding various models on different platforms (iOS, Android, Windows, BlackBerry, etc.). On top of that, new devices are hitting the market so often that it becomes impossible to cover all the major devices.

Multifaceted user interfaces: The user interface in desktop and browser based applications is typically limited to the mouse and keyboard. Mobile applications on the other hand are trying to make user interfaces as fluid as possible.

Application types – Web, native or hybrid: Native apps are created for use on a platform like a mobile or a tablet. Mobile Web apps are server-side apps to access websites on mobiles using different browsers like Chrome, Firefox, etc, by connecting to a mobile network or Wi-Fi network. Hybrid apps are a combination of native and Web apps. They run on devices or offline, and are written using HTML5 and CSS.

Agile development: Mobile apps are installed, uninstalled, or updated more regularly than desktop applications. As a tester in the mobile world, you have to be on top of the changes in the next revision of any OS or platform, and be clear about how these might affect an application.

Network: Web app testing is done mainly from one place with the help of LAN or Wi-Fi. Mobile applications rely on the network – the tester needs to test how the application responds to 3G, 4G, weak signals, no signals, powerful signals, when switching from cellular to wireless and vice

versa, or when the user is moving at different speeds -- all these variations can affect how the application behaves.

Important aspects of mobile and Web app testing

Mobile app testing

Operation: This relates to how apps interact with the phone, and how they react when any request is sent. This mainly depends on two factors.

- Connectivity: How do apps behave when there is no network or when in air plane mode?
- Interruptions: How do apps behave when there is some interruption like an incoming call, battery status indicator, etc.

Functionality: This is the core functionality required for an app to work properly. For example, does your app need Internet connectivity? What happens to an app when a user performs an action that the app is not designed to handle?

Installation: This is about whether the app is downloadable to all devices. Can it be installed and uninstalled properly without any error?

Versatility: This relates to how an app behaves in different situations or on different platforms. For example, how does it behave on a device with a different screen resolution, on a different mobile OS, etc.

Data security: This is the most important aspect of mobile testing, during which the tester checks what data the app saves, collects, processes and sends. The app needs to work well through all these tasks.

Web app testing

Link/URL testing: This relates to checking whether the correct link is opening and whether this happens properly.

Cookies testing: Testers check how the app behaves when enabling and disabling cookies.

Functional testing: During this process, testers check what core functionality the app requires to work properly, and how this is implemented.

Usability testing: System navigation is checked in this part. For example, how does the app respond when clicking on any link, do the correct responses emerge, etc.

Performance and security: Like in mobile testing, here too the tester tracks the risk of leaking private data. For example, what data is the app asking for and what is it doing with that data?

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